

## Limitations for GFC 700 Autopilot System:

Autopilot / Yaw Damper DISC during take-off and landing.

Do not use AP during single engine operation.

Maximum speed for autopilot operation is 180 KIAS.

Minimum speed for autopilot operation is 90 KIAS.

### Minimum Altitude for Autopilot Operation:

Cruise, Climb, Descent and Maneuvering : 800 feet AGL

Approach : 200 feet AGL

Departure : 200 feet AGL

|                  | Airspeed                          |                                              | KIAS     | Remarks                                                                    |
|------------------|-----------------------------------|----------------------------------------------|----------|----------------------------------------------------------------------------|
| V <sub>O</sub>   | Operating maneuvering speed       | above 1800 kg (3968 lb)                      | 122 KIAS | Do not make full or abrupt control surface movement above this speed.      |
|                  |                                   | above 1700 kg (3748 lb) to 1800 kg (3968 lb) | 119 KIAS |                                                                            |
|                  |                                   | up to 1700 kg (3748 lb)                      | 112 KIAS |                                                                            |
| V <sub>FE</sub>  | Max. flaps extended speed         | LDG                                          | 113 KIAS | Do not exceed these speeds with the given flap setting.                    |
|                  |                                   | APP                                          | 133 KIAS |                                                                            |
| V <sub>LO</sub>  | Max. landing gear operating speed | Extension V <sub>LOE</sub>                   | 188 KIAS | Do not operate the landing gear above this speed.                          |
|                  |                                   | Retraction V <sub>LOR</sub>                  | 152 KIAS |                                                                            |
| V <sub>LE</sub>  | Max. landing gear extended speed  |                                              | 188 KIAS | Do not exceed this speed with the landing gear extended.                   |
| V <sub>MCA</sub> | Minimum control speed airborne    | APP                                          | 68 KIAS  | With one engine inoperative, keep airspeed above this limit.               |
|                  |                                   | UP                                           | 71 KIAS  |                                                                            |
| V <sub>NO</sub>  | Max. structural cruising speed    |                                              | 151 KIAS | Do not exceed this speed except in smooth air, and then only with caution. |
| V <sub>NE</sub>  | Never exceed speed in smooth air  |                                              | 188 KIAS | Do not exceed this speed in any operation.                                 |

|                                                                                                                    | FLAPS | up to 1900 kg<br>(4189 lb) | above 1900 kg<br>(4189 lb) |
|--------------------------------------------------------------------------------------------------------------------|-------|----------------------------|----------------------------|
| Airspeed for rotation (take-off run, $V_R$ )                                                                       | UP    | min. 76 KIAS               | min. 76 KIAS               |
|                                                                                                                    | APP   | min. 71 KIAS               | min. 74 KIAS               |
| Airspeed for take-off climb<br>(best rate-of-climb speed $V_Y$ )                                                   | UP    | min. 90 KIAS               | min. 92 KIAS               |
| Airspeed for take-off climb<br>(best angle-of-climb speed $V_X$ )                                                  | APP   | min. 77 KIAS               | min. 79 KIAS               |
| Airspeed for best rate-of-climb ( $V_Y$ )                                                                          | UP    | 90 KIAS                    | 92 KIAS                    |
|                                                                                                                    | APP   | 85 KIAS                    | 85 KIAS                    |
| Airspeed for cruise climb                                                                                          | UP    | min. 90 KIAS               | min. 92 KIAS               |
| Reference landing approach speed                                                                                   | UP    | 86 KIAS                    | 92 KIAS                    |
|                                                                                                                    | APP   | min. 84 KIAS               | min. 88 KIAS               |
| Final approach speed                                                                                               | LDG   | min. 84 KIAS               | min. 86 KIAS               |
| Minimum speed during go around                                                                                     | UP    | min. 90 KIAS               | min. 90 KIAS               |
| Max. structural cruising speed<br>Do not exceed this speed except<br>in smooth air, and then only with<br>caution. | UP    | 151 KIAS                   | 151 KIAS                   |

| 1510 kg<br>(3329 lb) |       | Bank Angle |      |      |      |      |      |      |      |
|----------------------|-------|------------|------|------|------|------|------|------|------|
|                      |       | 0°         |      | 30°  |      | 45°  |      | 60°  |      |
| Gear                 | Flaps | KIAS       | KCAS | KIAS | KCAS | KIAS | KCAS | KIAS | KCAS |
| UP                   | UP    | 61         | 59   | 66   | 64   | 72   | 71   | 84   | 84   |
| DOWN                 | APP   | 58         | 57   | 63   | 62   | 69   | 68   | 81   | 81   |
| DOWN                 | LDG   | 54         | 54   | 60   | 59   | 67   | 65   | 79   | 77   |

| 1700 kg<br>(3748 lb) |       | Bank Angle |      |      |      |      |      |      |      |
|----------------------|-------|------------|------|------|------|------|------|------|------|
|                      |       | 0°         |      | 30°  |      | 45°  |      | 60°  |      |
| Gear                 | Flaps | KIAS       | KCAS | KIAS | KCAS | KIAS | KCAS | KIAS | KCAS |
| UP                   | UP    | 64         | 62   | 68   | 67   | 75   | 74   | 88   | 88   |
| DOWN                 | APP   | 62         | 61   | 66   | 66   | 73   | 73   | 86   | 86   |
| DOWN                 | LDG   | 59         | 58   | 64   | 62   | 71   | 69   | 86   | 82   |

| 1900 kg<br>(4189 lb) |       | Bank Angle |      |      |      |      |      |      |      |
|----------------------|-------|------------|------|------|------|------|------|------|------|
|                      |       | 0°         |      | 30°  |      | 45°  |      | 60°  |      |
| Gear                 | Flaps | KIAS       | KCAS | KIAS | KCAS | KIAS | KCAS | KIAS | KCAS |
| UP                   | UP    | 68         | 66   | 72   | 71   | 79   | 78   | 93   | 93   |
| DOWN                 | APP   | 65         | 64   | 70   | 69   | 77   | 76   | 90   | 91   |
| DOWN                 | LDG   | 62         | 61   | 67   | 66   | 75   | 73   | 89   | 86   |